

Monte Carlo Notes

M08: Lecture 4 (Pages 8-14)

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Importance sampling

- Sometimes classical MC is impossible because we can't sample from the target distribution $p(x)$
- Instead, we sample from a simpler distribution $q(x)$, the instrument density, which now becomes the target distribution

$$\begin{aligned} E_p[f(X)] &= \int f(x)p(x)dx \\ &= \int f(x)\frac{p(x)q(x)}{q(x)}dx \\ &= \int z(x)q(x)dx \\ &= E_q[z(x)] \\ &\approx \frac{1}{n} \sum_{i=1}^n z(x_i) \quad \text{By LLN} \end{aligned}$$

$$\begin{aligned} \text{where: } x_1, \dots, x_n &\stackrel{i.i.d}{\sim} q(x) \\ &\approx \frac{1}{n} \sum_{i=1}^n f(x_i) \frac{p(x_i)}{q(x_i)} \\ &\quad \quad \quad W_i = \text{Weights, or} \\ &\quad \quad \quad \text{likelihood ratio!} \\ &\approx \frac{\sum_{i=1}^n f(x_i) \frac{p(x_i)}{q(x_i)}}{\sum_{i=1}^n \frac{p(x_i)}{q(x_i)}} \end{aligned}$$

- The last equation is required when we don't know the normalizing constant of $p(x)$!
- Importance sampling also helps us to decrease the MC error (a problem listed above)!

Elementary Markov chain theory

- A series of random variables X_t is a stochastic process
- Markov property:

- $\Pr(X_{n+1} = j | X_n = i, \dots, X_0) = \Pr(X_{n+1} = j | X_n = i)$
- One-step transition probability:
 - $p_{ij} = \Pr(X_1 = j | X_0 = i) \quad \sum_j p_{ij}, p_{ij} \geq 0 \text{ for all } i, j$
- X_n is a homogenous Markov chain if the transition probabilities are the same for all transitions
 - Inhomogenous otherwise
- Collect the into a **transition probability matrix** $\mathbf{P}^{(n)} = \{p_{ij}^{(n)}\}$ where (n) refers to the number of steps in the transition
 - $\mathbf{P}^{(n)} = \mathbf{P}^n$ i.e. n -step transition probabilities can be obtained by repeatedly multiplying the transition probability matrix by itself
- A Markov chain is fully specified by its transition probability matrix $\mathbf{P}^{(n)}$ and its initial distribution $\nu^{(n)}$
 - $\nu^{(n)} = \mathbf{P}^n \nu^{(0)}$ given $\nu^{(0)} = \nu_1^{(0)}, \dots, \nu_s^{(0)}$ where $\nu_i^{(0)} = \Pr(X_0 = i)$
 - Allows the marginal distribution of X_n to be easily computed
- In a discrete distribution, the time spent in each state is distributed according to a Geometric distribution as it's the time until the move to the next (the first event)
 - In a continuous distribution, the geometric distribution becomes an exponential distribution

Behavior of Markov chains

- Irreducible
 - If you can for from anywhere to anywhere in a finite number of steps with a non-0 probability
 - * $p_{ij}^{(n)} > 0$ where $n > 0$
 - * I.e. the graph is connected so can't get stuck in a state
- Stationary distribution and long term behavior
 - If we have a special vector of probability distributions (usually defined as a column vector that is then *transposed* for the matrix multiplication) on state space E that satisfies $\pi^T = \pi^T \mathbf{P}$ (**global balance**), it is called a stationary (or **equilibrium**) distribution of the corresponding homogeneous Markov chain
 - * What this means is that at each time step, the probability distributions at each value of the state space ($\pi_1, \dots, \pi_n$) are the same as the previous time step!
 - Doesn't mean that underlying observations don't change, just the overall distribution at the value of the state space
 - * Means that the Markov chain becomes very predictable
 - Hard to solve unless in a small finite state space
- Reversible Markov chain - detailed balance equations
 - $\pi_i p_{ij} = \pi_j p_{ji}$ for all i, j then π is a stationary distribution (detailed balance equation)
 - * Balance preserved between all pairs (**global balance**)
 - When this condition is met, they are often called **reversible Markov chains**
- Using the lab example of the Ehrenfest diffusion model, we see the following transition probabilities, and how we end up in a stationary distribution

Markov chains

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